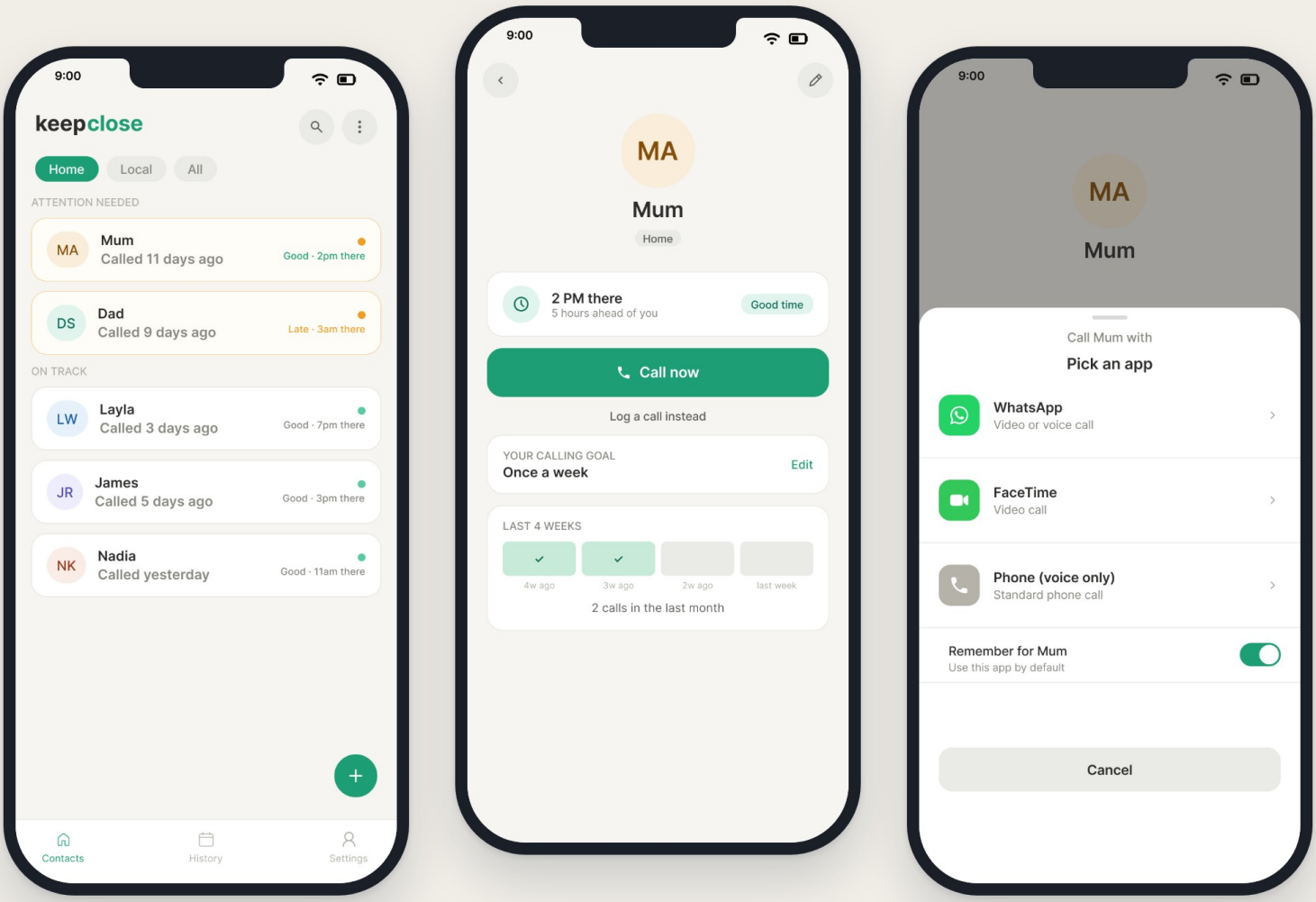
UX & PRODUCT DESIGN CASE STUDY · AYOUB ALDAWOOD

keepclose

BSc dissertation project · Newcastle University, UK

A mobile app that helps international students mitigate homesickness and loneliness by supporting routine-based calls with family and friends.

[View prototype (Figma)]



ROLE

Research, design, prototyping and evaluation

METHODS

Survey (50 students); scenario-based walkthroughs (10 students)

TOOLS

Figma, Google Forms

ABOUT THIS CASE STUDY

This case study summarizes my 79–page BSc dissertation at Newcastle University (2026), titled “Supporting Homesickness and Loneliness in International Students: Designing a Wellbeing-Maintenance Calling Tool”, supervised by Dr. Nick Taylor. The full dissertation is available on request.

1. Overview

International students often experience homesickness and loneliness, and calls with family and friends are one of the everyday ways they cope. Yet regular calls are hard to keep up: time zones, study demands and daily responsibilities push them back until a postponed call becomes a gap of weeks. keepclose helps students keep those calls going. It works alongside the calling apps they already use, such as WhatsApp and FaceTime, and adds what those apps lack: a calling goal for each person, a clear sense of whom to call and when, and a calm view of calling patterns.

- **Problem:** existing communication apps enable contact, but they don't support maintaining healthy calling routines over time.
- **Research:** a survey of 50 international students found that 90% experience connection drift (calls with someone important happening less often than they'd like), mainly because of study workload, time-zone differences and daily responsibilities. It also revealed the central design risk: gentle reminders were the most wanted feature, and too many reminders the biggest deterrent, both at 74%.
- **Design:** a high-fidelity Figma prototype with seven screens covering three core journeys: adding a contact, noticing connection drift then calling, and reviewing calling patterns. It deliberately avoids streaks, scores and automatic tracking.
- **Evaluation:** scenario-based walkthroughs with 10 international students. All ten completed every journey with no failed steps, and the design's restraint drew the strongest endorsement.
- **Next:** warmer wording for two key labels, editable Circles, a clearer History screen, visible privacy reassurance and more calling apps (section 7).

2. The design

keepclose supports a simple routine: students set how often they'd like to call each person, see at a glance whom to call and whether it's a good time where that person is, start the call in the app they already use, and look back on their calling patterns. The prototype covers it in three journeys: adding a contact (Figure 1), noticing connection drift then calling (Figure 2), and reviewing calling patterns (Figure 3). Settings appears in the tab bar but wasn't built, as no journey needed it.

Avoided on purpose

User research was as clear about what would drive students away as about what they wanted, so keepclose is defined as much by what it avoids as by what it offers.

What keepclose deliberately avoids

- **Streaks:** 44% said streak pressure would put them off.
- **Scores, percentages and rankings,** and any comparison between people.
- **Badges and achievements.**
- **Automatic access** to the phone: adding contacts and logging calls are both done by hand; only 26% wanted the app to read their phone's call history.
- **Frequent nudges:** 74% named too many reminders as a deterrent.
- **Alarming colors:** soft amber, never red, marks the people who need attention.

Adding someone to keep in touch with

Adding a contact is where the routine begins. The form asks only for what keepclose needs, and makes the student's calling intention explicit from the start: a vague “I should call more” becomes “once a week”.

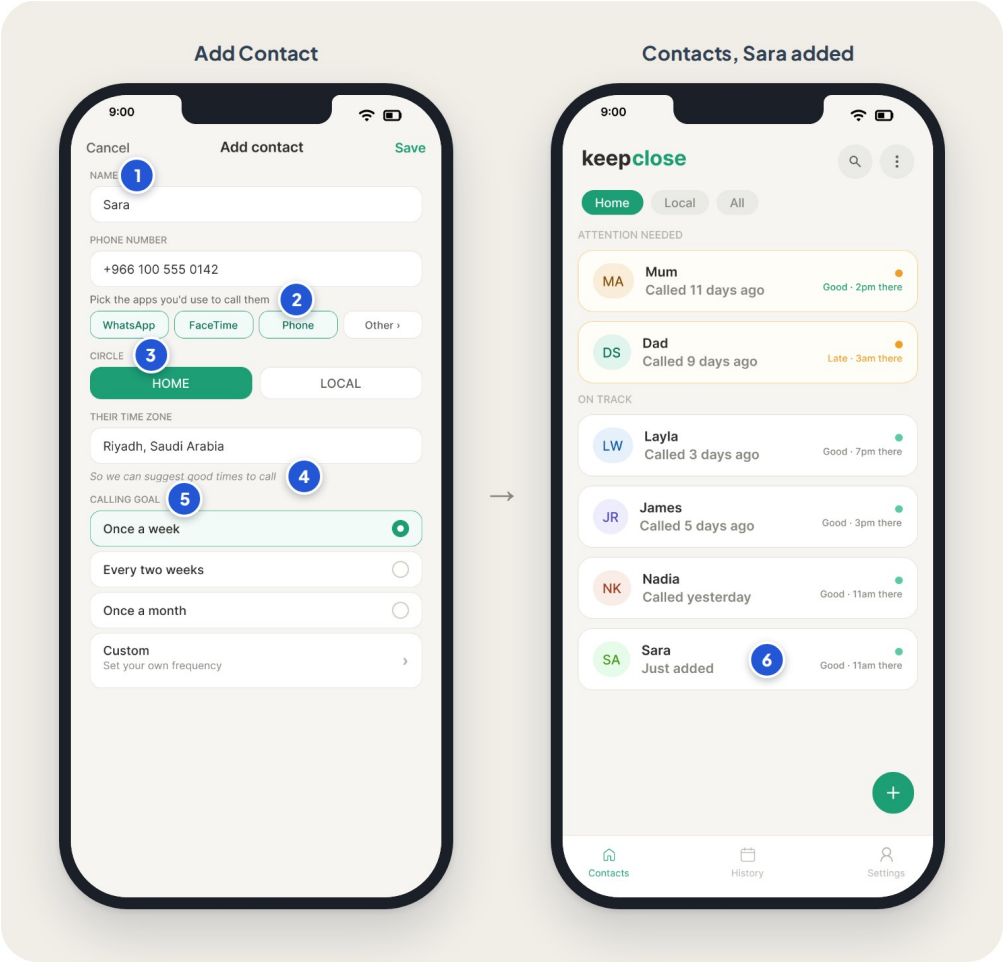


Figure 1. Adding a contact: the completed Add Contact form (left) and Contacts with Sara added (right).

1. Contacts are entered by hand, with only the details the app needs. Nothing is read from the phone's call history or address book, by design: 40% of students worried about feeling monitored, and only 26% wanted automatic access.
2. The apps chosen here are the ones the call sheet (Figure 2) will offer for this person.
3. Each contact belongs to a Circle: “Home” for people back in the student's home country, or “Local” for people in the country where they study. The two default Circles appear as tabs on Contacts.
4. Each field explains why it's needed: the time zone is “so we can suggest good times to call”.
5. The calling goal is set for each person, with three presets and “Custom”: one tap to set up, with the choice left to the student.
6. Back on Contacts, Sara appears under “On Track”, with her local time already shown.

Knowing who to call, and when

This is the core journey. Contacts shows each person's local time and whether it's a good time to call, and anyone not called within the student's calling goal moves to the top. From there, “Call now” is one tap away.

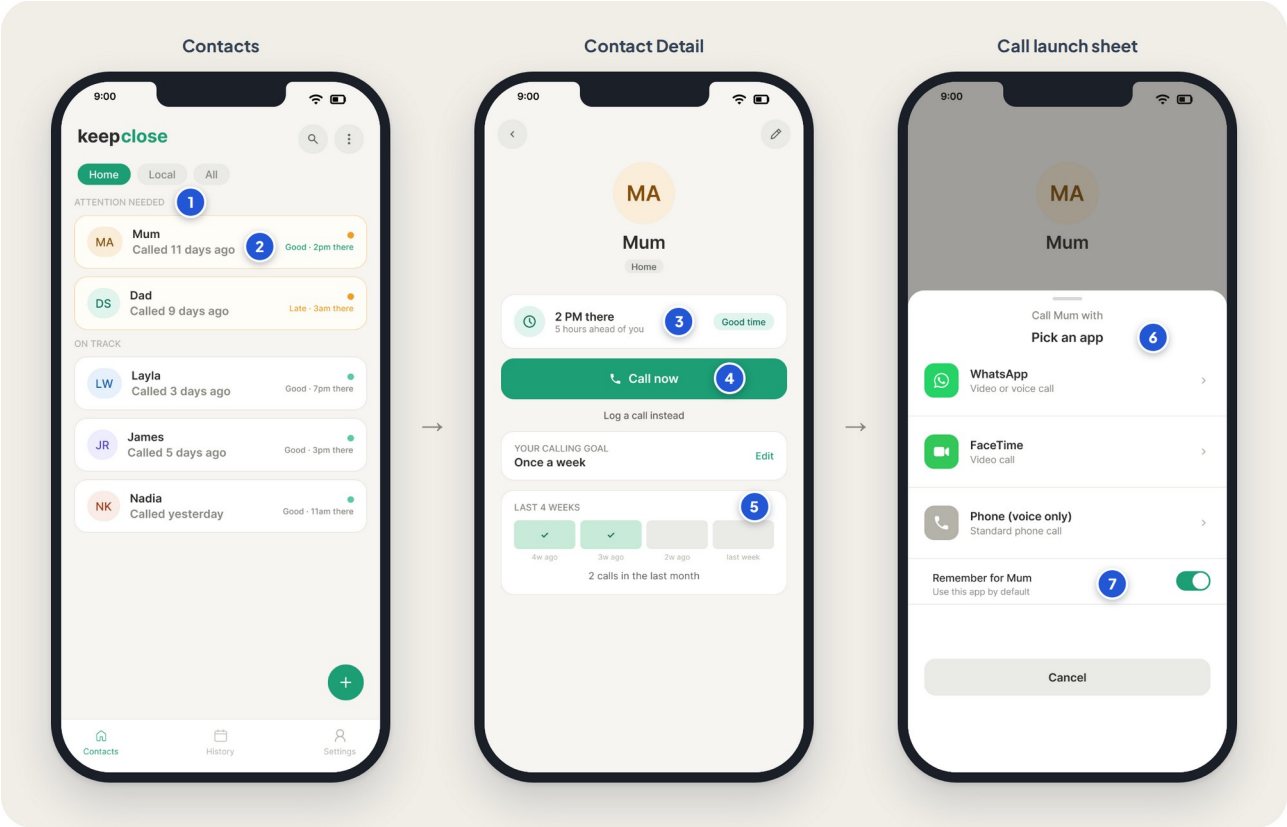
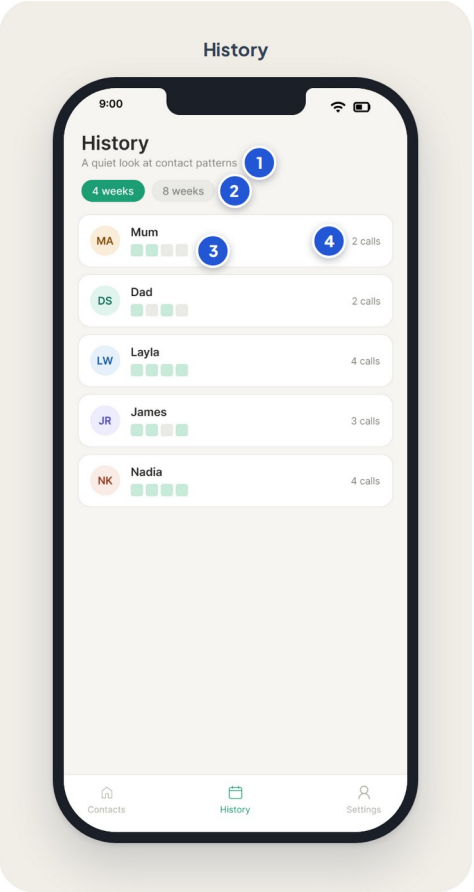


Figure 2. Noticing drift then calling: Contacts (left), Contact Detail for Mum (middle), and the call launch sheet (right).

1. Anyone not called within the student's calling goal moves to the top under “Attention Needed”, marked by a soft amber dot rather than a red badge, so it reads as a gentle prompt, not a warning.
2. Every card shows the person's local time and whether it's a good time to call, so students can decide from the list itself.
3. The time-zone helper turns the time difference into a clear answer: “2 PM there” and “Good time”.
4. “Call now” is the primary action, one tap from opening keepclose. “Log a call instead” is a quieter, secondary action for recording a call by hand, so History stays up to date.
5. The last four weeks appear as ticked or empty tiles with one plain sentence. There's no score, streak or percentage, so a quiet week isn't framed as failure.
6. “Call now” opens a choice of the apps the student selected for Mum, such as WhatsApp and FaceTime. keepclose is a layer on top of them, not a replacement.
7. “Remember for Mum” makes that app the default, so the next call is quicker.

Reflecting on calling patterns

History offers “a quiet look at contact patterns”: it builds awareness of calling habits over time without turning them into a score or streak to chase and worry about.



1. The title and subtitle frame the screen as reflection rather than evaluation.
2. The only control is the range: “4 weeks” or “8 weeks”.
3. One tile per week, filled if there was a call: a pattern to glance at, not a score to chase.
4. A neutral count for each person, with no streaks, rankings, totals or comparisons between people.

Figure 3. History, showing calling patterns over the last four weeks.

How I got there

The problem, user research, requirements and evaluation behind the design.

3. The problem

Homesickness and loneliness are well-documented parts of the international student experience (Sawir et al., 2008; Zheng et al., 2023), and calls with family and friends are one of the everyday ways students cope (Kelly et al., 2021). But regular contact drifts. Study demands, time zones and the effort of starting a call push calls back, and what begins as a postponement becomes a gap. Existing communication apps make the call itself easy, but they assume the user already knows whom to call, when, and how often.

Neighboring products don't fill this gap. Habit trackers such as Streaks and Duolingo motivate through streaks and scores, a pressure that doesn't belong in relationships. Personal CRM apps such as UpHabit share the structure of contact goals and reminders, but treat people as a network to manage. Calm apps such as Headspace have the right tone, but address mood rather than relationships. keepclose combines the structure of the second with the tone of the third, without the mechanics of the first.

4. User research

I surveyed 50 international students through an anonymous online form about how they keep in touch, when and why contact drifts, and which features would help or put them off. Calling clearly matters: 84% rated voice and video calls more helpful than messaging for feeling connected. Drift is common: 90% experience it at least sometimes, and 68% respond by switching to messaging rather than rescheduling the call. Its causes are mostly practical (Figure 4).

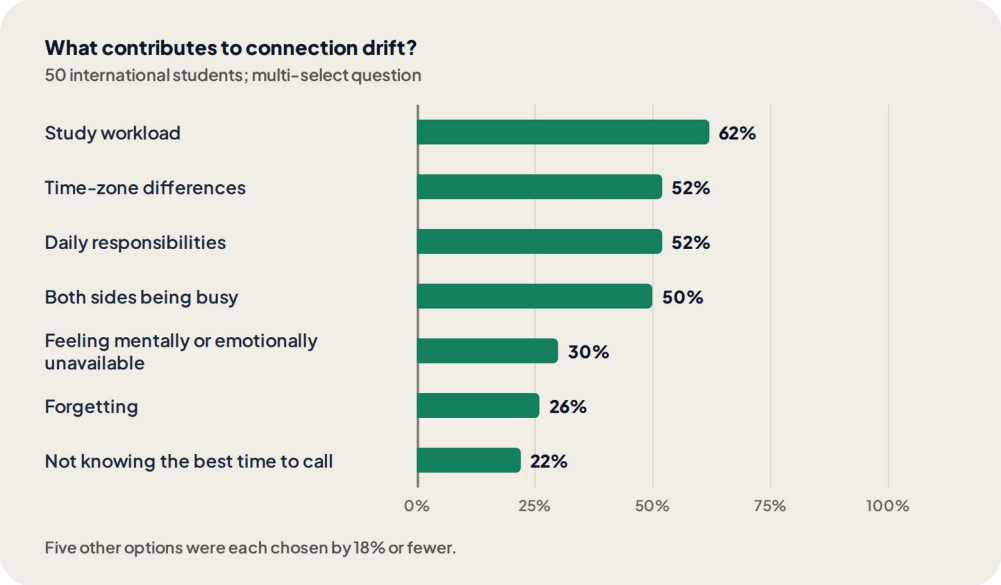


Figure 4. What contributes to connection drift (50 international students; multi-select question).

The feature questions revealed the central design risk (Figure 5): gentle reminders were the most wanted feature, and too many reminders the most common reason to avoid an app like this, both at 74%. A reminder that helps at one frequency becomes intolerable at another. Streak pressure (44%) and feeling monitored (40%) followed, and 84% preferred a gentle or neutral tone.

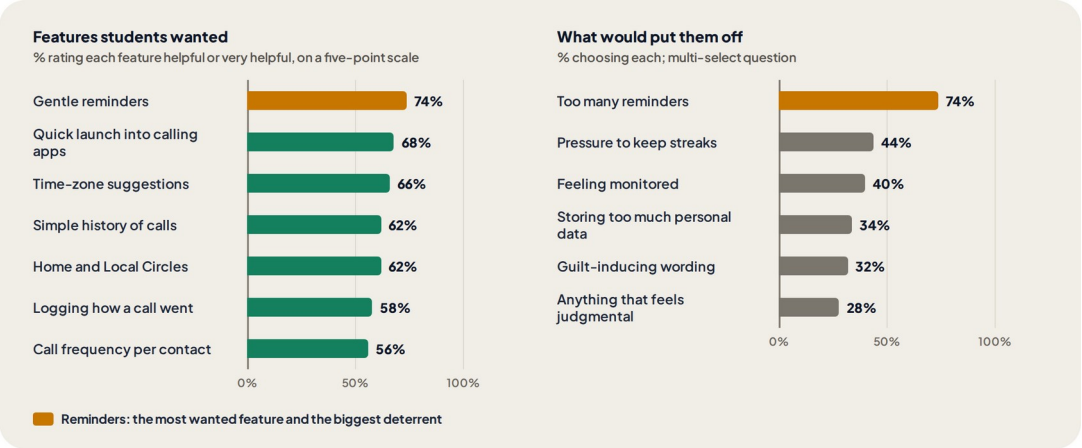


Figure 5. Features rated helpful (left) and what would make the app less appealing (right). Reminders top both lists, at 74%.

5. From findings to requirements

Following a human-centered design process (Sharp, Rogers and Preece, 2019), I translated the findings into seven functional and seven non-functional requirements, each traced to its survey evidence. Peters' (2023) wellbeing-supportive design guidelines served as the lens: support autonomy, and prefer low pressure to pressure, even where pressure works in the short term. Here, the non-functional requirements matter as much as the features: a judgmental tone or a hint of surveillance would undermine the app's purpose (Table 1).

What students told me	What the design does
Time-zone differences were the second most common cause of connection drift (52%).	Every contact card shows local time and status, and each contact's page answers “Is now a good time?”
When calls slip, 68% fall back on messaging, the easier option.	Calling is almost as quick as messaging: “Call now” is one tap from opening keepclose, and the call opens in the student's chosen app, such as WhatsApp, FaceTime or the phone.
74% wanted gentle reminders, yet 74% said too many reminders would put them off.	People not called within their goal rise to the top under a soft “Attention Needed” heading, and reminder notifications are specified to stay rare: at most once a week per contact by default, adjustable by the student.
44% cited streak pressure, and 26% feel guilty when contact slips.	No streaks, scores, badges or comparisons anywhere.
40% worried about feeling monitored; only 26% wanted the app to have automatic access to their call history.	Nothing is read from the phone: adding contacts and logging calls are both done by hand.
84% preferred a gentle or neutral tone; 32% objected to guilt-inducing wording.	Calm copy, cream backgrounds, one muted green, and amber instead of red.

What students told me	What the design does
56% found a per-contact calling frequency helpful; one student warned that a fixed "Home" label can hurt those who now feel at home abroad too.	A calling goal for each person, and "Home" and "Local" Circles with labels designed to be editable.

Table 1. How the main survey findings shaped the design.

Structure

The requirements became five functional areas (Figure 6). The app opens on people rather than a dashboard, because the people students most regularly keep in touch with are a small circle, mostly family and close friends (parents 94%, siblings 70%, close friends from home 52%). Call launch is a sheet that slides up rather than a new page, signaling that keepclose hands the call over to another app. Adding a contact sits behind a + button on Contacts, one tap away.

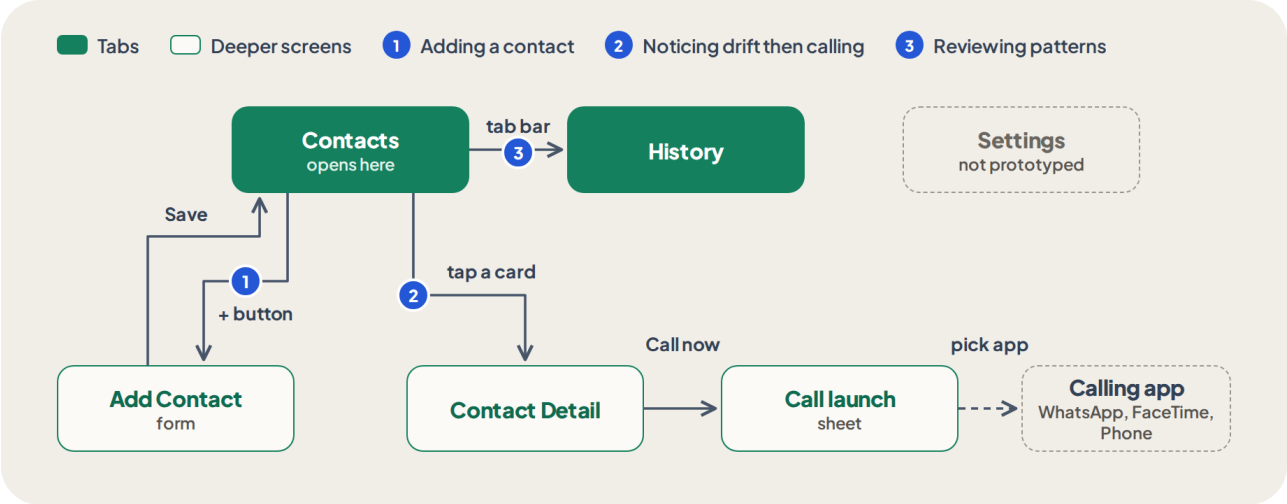


Figure 6. App structure and the three journeys. Settings appears in the tab bar but wasn't built, as no journey needed it.

Alternatives and fidelity

Before building in Figma, I weighed and rejected alternatives in writing (e.g., a dashboard-first vs. contact-first home, and an achievement-style vs. neutral history visualization). I built and evaluated the prototype at high fidelity because tone and visual restraint were part of what I was testing, and both are largely invisible in gray wireframes or low-fidelity prototypes.

6. Evaluation

I evaluated the prototype through scenario-based walkthroughs rather than a heuristic review or timed measures such as the System Usability Scale. Those methods judge an app like a productivity tool, while this design also had to feel right: calm rather than judgmental, and supportive rather than surveilling.

Method

- **Participants:** 10 international students, from foundation level to PhD, abroad for between 3 months and 4 years, with time-zone differences of 1 to 9 hours between where they studied and home.
- **Sessions:** 25 to 34 minutes each; eight in person and two by video call.

- **Scenarios:** one per journey, each set in a realistic situation: your sister has just moved abroad; it's been a while since you called your mum; it's the end of the month and you want a sense of how you've kept in touch.
- **Data:** the outcome of five steps per scenario (completed, needed a hint, or failed), answers to open questions after each scenario, and session notes, analyzed through light thematic analysis (Braun and Clarke, 2006).
- **Ethics:** participation was voluntary, with informed consent and anonymous data; no identifying details were collected, and sessions weren't recorded, keeping the data minimal on a sensitive topic. The study ran under Newcastle University's ethical approval for low-risk undergraduate projects, and participants are referred to as P01 to P10.

Task completion

No participant failed a step, and 138 of 150 steps were completed without help (Table 2). Most hints were needed on History, which pointed to the clearest usability fix: the screen lacks a legend.

Journey	Without help	With a hint	Failed	Where hints were needed
Adding a contact	49 of 50	1	0	One didn't notice the + button at first.
Noticing drift then calling	47 of 50	3	0	Two didn't use the time-zone helper to decide; one expected "Call now" to call straight away rather than offer a choice of apps.
Reviewing patterns	42 of 50	8	0	Five needed a moment to work out the week tiles; three missed the "4 weeks" and "8 weeks" toggle.

Table 2. Task completion across 150 observed steps (10 participants, 3 scenarios, 5 steps each).

What worked

10 of 10 **No streaks or scores.** Endorsed by every participant. Several contrasted History with fitness and productivity apps, and one said scores would make it feel "like Duolingo".

10 of 10 **A layer, not a replacement.** No one wanted another calling app. One participant summed up the value as deciding whom to contact, then quickly launching the right app.

9 of 10 **Calling goals.** Setting how often to call gave structure to good intentions, though some found the word "goal" too directive and didn't want goals for friendships.

8 of 10 **The time-zone helper.** The most warmly endorsed feature, valued less as information than as a nudge to act: "The time-zone part removes the excuse of not knowing" (P07).

What needs work

8 of 10 **The "Attention Needed" heading.** Everyone understood what it did, but most heard obligation in it: "Relationships are not overdue library books" (P08).

6 of 10 **The "Home" and "Local" Circles.** Too binary for students whose lives span more than one country: "My home is not only where my passport is" (P06).

5 of 10 **The History tiles.** Half needed a moment to work out that each tile is a week, and two asked for a legend or explanation.

4 of 10 **How History feels.** The calm design softened the weight of seeing one's own pattern but didn't remove it: three felt mild guilt at empty weeks, and one felt judged and wanted to hide the screen.

3 of 10 **Privacy.** Three raised it unprompted. The design already avoids tracking, but nothing on screen says so.

Limitations

Overall, the evaluation offers qualified support for the design: participants could use keepclose, and they recognized and endorsed its restraint without being told about it. Its limits are those of a first study:

- **One session, imagined situations:** a 30–minute walkthrough shows how a design lands at first contact, not whether it supports wellbeing over weeks of real use.
- **Small, self-selected samples:** 50 students surveyed and 10 tested, enough to shape and check the design, but too few to draw comparisons between groups, such as newly arrived vs. longer-term students.
- **Reminders untested:** notifications were specified but not prototyped, so how often reminders should arrive couldn't be tested.

7. Next steps

- **Warmer wording:** replace "Attention Needed" and "Calling goal" with gentler phrasing drawn from participants' suggestions, such as "ideal rhythm", and offer the option of having no goal at all.
- **Editable Circles:** let students rename the "Home" and "Local" Circles, or create new ones.
- **History:** add a small legend for the tiles, let a tap on a week show what happened, and offer an option to hide the screen.
- **Visible privacy:** state plainly at first use that keepclose never reads calls, messages or contacts.
- **More calling apps:** add regional apps such as WeChat to the default choices.

8. Reflection

- **Restraint can be the design.** The most consistent praise was for what keepclose avoids. In a domain this personal, what an app refuses to do matters as much as what it offers.
- **Wording is a design surface, not polish.** Reminders were both the most wanted feature and the biggest deterrent, and "Attention Needed" worked as a function but failed as words: wording decided how each was received.
- **Protection has to be visible.** The design already avoided tracking, yet participants still worried about it. A safeguard people can't see doesn't reassure them.
- **Methods shape findings.** Walkthroughs captured how the design felt, which timed tasks would have missed, but a single session can't show what happens over weeks.

Selected references

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